EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

General Properties of the d-Block Elements (Physical Properties, Atomic and Ionic Size, density)

1. Coinage metals show the properties of

(A) typical elements (B) normal elements

(C) inner-transition elements (D) transition element

CDF084

2. **Assertion**: Zinc does not show characteristic properties of transition metals.

Reason : In zinc outermost shell is completely filled.

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

CDF089

Metallic bond strength, melting point, enthalpy of atomisation

3. The d-block element which is a liquid at room temperature, having high specific heat, less reactivity than hydrogen and its chloride (MX_2) is volatile on heating is

(A) Cu (B) Hg (C) Ce (D) Pm

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CDF083

4. **Assertion**: Tungsten has a very high melting point.

Reason : Tungsten is a covalent compound.

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

CDF090

Variable Oxidation States, Stability of higher O.S.

5. Transition elements are usually characterised by variable oxidation states but Zn does not show this property because of

(A) completion of np-orbitals (B) completion of (n–1)d orbitals

(C) completion of ns-orbitals (D) inert pair effect

CDF082

6. The higher oxidation states of transition elements are found to be in the combination with A and B, which are

(A) F, O (B) O, N (C) O, Cl (D) F, Cl

CDF085

For More Join: @IITJEE_Advanced

7. The electrons which take part in order to exhibit variable oxidation states by transition metals are

(A) ns only (B) (n–1)d only

(C) ns and (n–1)d only but not np (D) (n–1)d and np only but not ns

CDF086

8. **Assertion**: The highest oxidation state of chromium in its compounds is +6.

Reason : Chromium atom has only six electrons in ns and (n–1) d orbitals.

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

CDF088

Complex formation, Catalytic Properties

9. Transition elements having more tendency to form complex than representative elements (s and p-block elements) due to -

(A) availability of d-orbitals for bonding

(B) variable oxidation states are not shown by transition elements

(C) all electrons are paired in d-orbitals

(D) *f*-orbitals are available for bonding

CDF081

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Alloys formation

10. Meter scales are made-up of alloy

(A) invar (B) stainless steel (C) electron (D) magnalium

CDF087

Some Important Compounds of Transition Elements and their (Physical and Chemical) properties ($K_2Cr_2O_7$, K_2CrO_4, $KMnO_4$, K_2MnO_4)

11. Solution of $MnO_4^{\ominus}$ is purple-coloured due to

(A) d-d-transition

(B) charge transfer from O to Mn

(C) due to both d-d-transition and charge transfer

(D) none of these

CDF092

12. During estimation of oxalic acid Vs $KMnO_4$, self-indicator is

(A) $KMnO_4$ (B) oxalic acid (C) K_2SO_4 (D) $MnSO_4$

CDF093

13. Acidified chromic acid + $H_2O_2 \xrightarrow{\text{Org.solvent}} \underset{\text{(blue colour)}}{X} + Y$,

X and Y are :

(A) CrO_5 and H_2O (B) Cr_2O_3 and H_2O (C) CrO_2 and H_2O (D) CrO and H_2O

CDF094

For More Join: @IITJEE_Advanced

14. $KMnO_4 + HCl \longrightarrow H_2O + X(g)$, X is a
(acidified)

(A) red liquid
(B) violet gas
(C) greenish yellow gas
(D) yellow-brown gas

CDF095

15. **Assertion**: $KMnO_4$ is purple in colour due to charge transfer.

Reason : In $MnO_4^{\ominus}$, there is no electron present in d-orbitals of manganese.

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)
(B) if both (A) and (R) are true but (R) is not correct explanation of (A)
(C) if (A) is true but (R) is false
(D) if (A) is false and (R) is true

CDF096

16. **Assertion**: K_2CrO_4 has yellow colour due to charge transfer.

Reason : $CrO_4^{2\ominus}$ ion is tetrahedral in shape.

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

Bankers Academy JEE

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)
(B) if both (A) and (R) are true but (R) is not correct explanation of (A)
(C) if (A) is true but (R) is false
(D) if (A) is false and (R) is true

CDF097

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

General Properties of the d-Block Elements (Physical Properties, Atomic and Ionic Size, density)

1. The lanthanide contraction is responsible for the fact that
(A) Zr and H*f* have same atomic sizes (B) Zr and H*f* have same properties
(C) Zr and H*f* have different atomic sizes (D) Zr and H*f* have different properties

CDF104

Metallic bond strength, melting point, enthalpy of atomisation

2. Mercury is a liquid at 0°C because of
(A) very high ionisation energy (B) weak metallic bonds
(C) high heat of hydration (D) high heat of sublimation

CDF099

Ionisation Enthalpies

3. The ionisation energies of transition elements are
(A) less than p-block elements (B) more than s-block elements
(C) less than s-block elements (D) more than p-block elements

CDF100

4. Amphoteric oxide(s) of Mn is/are
(A) MnO_2 (B) Mn_3O_4 (C) Mn_2O_7 (D) MnO

CDF105

Rankers Academy JEE

Variable Oxidation States, Stability of higher O.S.

5. The highest oxidation state among transition elements is
(A) + 7 by Mn (B) + 8 by Os (C) + 8 by Ru (D) + 7 by Fe

CDF101

Complex formation, Catalytic Properties

6. The catalytic activity of transition elements is related to their
(A) variable oxidation states (B) free surface valencies
(C) complex formation ability (D) magnetic moment

CDF102

Formation of interstitial compounds

7. Addition of non-metals like B and C to the interstitial sites of a transition metal results the metal
(A) of more duct ability (B) of less duct ability
(C) less malleable (D) of more hardness

CDF098

8. Interstitial compounds are formed by
(A) Co (B) Ni (C) Fe (D) Ca

CDF103

Some Important Compounds of Transition Elements and their (Physical and Chemical) properties ($K_2Cr_2O_7$, K_2CrO_4, $KMnO_4$, K_2MnO_4)

9. Acidified $KMnO_4$ can be decolourised by
(A) SO_2 (B) H_2O_2 (C) $FeSO_4$ (D) $Fe_2(SO_4)_3$

CDF106

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Complex formation, Catalytic Properties

Paragraph for Question No. 10 to 11

Transition metal and their compounds are used as catalysts in industry and in biological system. For example, in the Contact process, vanadium compounds in the +5 state (V_2O_5 or VO_3^-) are used to oxidise SO_2 to SO_3:

$$SO_2 + \frac{1}{2}O_2 \xrightarrow{V_2O_5} SO_3$$

It is thought that the actual oxidation process takes place in two stages. In the first step, $V^{5\oplus}$ in the presence of oxide ions converts SO_2 to SO_3. At the same time, $V^{5\oplus}$ is reduced to $V^{4\oplus}$.

$$2V^{5\oplus} + O^{2\ominus} + SO_2 \longrightarrow 2V^{4\oplus} + SO_3$$

In the second step, V^{5+} is regenerated from V^{4+} by oxygen :

$$2V^{4\oplus} + \frac{1}{2}O_2 \longrightarrow 2V^{5\oplus} + O^{2\ominus}$$

The overall process is, of course, the sum of these two steps:

$$SO_2 + \frac{1}{2}O_2 \longrightarrow SO_3$$

10. During the course of the reaction **:**

(A) Catalyst undergoes changes in oxidation state

(B) Catalyst increases the rate constant

(C) Catalyst is regenerated in its original form when the reactants form the products

(D) All are correct.

CDF107

11. Which of the following ion involved in the above process will show paramagnetism?

(A) $V^{5\oplus}$ (B) $V^{4\oplus}$ (C) $O^{2\ominus}$ (D) $VO_3^{\ominus}$

CDF108

Bankers Academy JEE

Formation of interstitial compounds

Comprehension # 12 to 14

Due to availability of vacant orbitals of sufficiently low energy, d-block elements form complexes, d-block elements have different properties such as- catalytic, magnetic, alloy formation, interstitial compounds formation. Interstitial compounds are those compounds in which small atoms like carbon and boron fits into interstices of d-block elements crystal. In interstitial compounds, there is no chemical bond formation takes place so, chemical properties remain almost same but physical properties may change.

12. Which of the property of interstitial compounds has the same behavior as that of the element -

(A) Malleability (B) Ductility

(C) Electrical conductance (D) Hardness

CDF109

13. Which of the following property gets decreased in interistitial compounds compared to that of the element -

(A) Malleability (B) Metallic lustre (C) Hardness (D) Density

CDF110

14. Select correct statement -

(A) Highest oxidation state of 3d-series is +8.

(B) Ni, Cu and Zn are not transition element.

(C) Ziglar natta catalyst contain vanadium.

(D) Aq. solution of $Cu^{2\oplus}$, $Fe^{3\oplus}$ and $Cr^{3\oplus}$ are blue, yellow and green respectively.

CDF111

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Electronic Configuration of the d-Block Elements

1. How many elements of 1st transition series have half-filled d-subshell ?

CDF113

2. Select the number of correct matches for the outermost electronic configuration for d-block elements.

Nb-$5s^1 4d^4$(group-5) W-$6s^2 5d^4$ (group-6)
Tc-$5s^1 4d^6$(group-7) Ru-$5s^1 4d^7$ (group-8)
Rh-$5s^1 4d^8$ (group-9) Pd-$5s^0 4d^{10}$(group-10)
Pt-$6s^1 5d^9$ (group-10) V-$4s^2 3d^3$ (group-5)

CDF115

Variable Oxidation States, Stability of higher O.S.

3. Find the number of species among the following which can exist.

TiO_2, $VO_2^{\oplus}$, $CrO_4^{2\ominus}$, $MnO_4^{\ominus}$, $FeO_4^{2\ominus}$, $VO^{2\oplus}$, $TiO^{2\oplus}$, Sc_2O_3, Mn_2O_7

CDF116

4. Find the number of halides which exist.

TiI_4, TiI_3, TiI_2, TiF_2, $TiBr_2$, $TiCl_2$, TiF_4, $TiCl_4$, $TiCl_3$

Rankers Academy JEE

CDF146

5. The number of halides that do not exist among the following.

CuF_2, $CuCl_2$, $CuBr_2$, CuI_2, CuCl, CuBr, Cu_2I_2, CuF

CDF118

6. Find the number of mixed oxides among the following :

Mn_3O_4, Fe_3O_4, Co_3O_4, Mn_2O_3, MnO_2, Fe_2O_3, NiO, CrO_2, CrO_3.

CDF123

7. Identify the number of amphoteric species amongst the following :

Mn_2O_7, CrO_3, V_2O_5, MnO_2, V_2O_3, CrO, Fe_2O_3

CDF124

8. Total number of transition metals which have only one oxidation state other than zero.

Fe, Cd, Zn, Ag, Au, Cu, Ni

CDF147

Magnetic Properties

9. Find the number of species from the following which has magnetic moment value of 1.73 B.M.

$Fe^{2\oplus}$, $Cu^{2\oplus}$, $Ni^{2\oplus}$, NO_2, $NO_2^{\ominus}$, $Ti^{3\oplus}$

CDF112

10. (i) Number of unpaired electrons in $Ti^{3\oplus}$ = x

(ii) Square of 'spin only' magnetic moment of $Cr^{3\oplus}$ = y determines, (y – x)

CDF148

For More Join: @IITJEE_Advanced

Complex formation, Catalytic Properties

11. What is the oxidation no. of metal in catalyst which converts SO_2 to SO_3 in contact process for the making of H_2SO_4 ?

CDF114

Formation of interstitial compounds

12. Number of the property gets increased in interstitial compounds compared to that of the element.
(i) Malleability (ii) Melting point (iii) Hardness (iv) density

CDF028

Some Important Compounds of Transition Elements and their (Physical and Chemical) properties ($K_2Cr_2O_7$, K_2CrO_4, $KMnO_4$, K_2MnO_4)

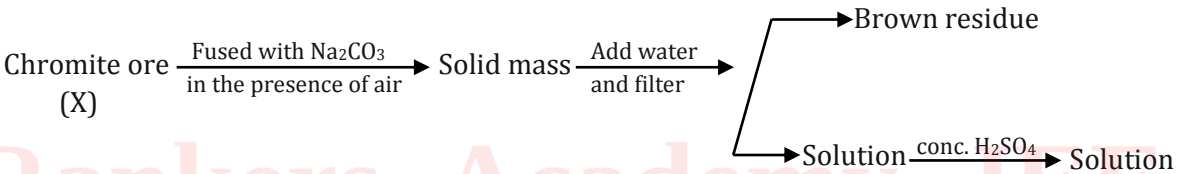
13. When mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc.H_2SO_4 then compound X is formed. What is the oxidation state of central atom of X.

CDF119

14. Find the total number of triangular faces in the geometry of $MnO_4^{\ominus}$ ion.

CDF120

15. Following steps are involved in manufacturing of potassium dichromate :
What is difference in the O.N. of Cr between X and Z ?



Rankers Academy JEE

CDF121

16. Find total no. of oxygen atom which are in –1 oxidation state in CrO_5.

CDF122

17. (i) Change in oxidation state of Cr when $K_2Cr_2O_7$ reacts with H_2S in acidic medium = x
(ii) Change in oxidation state of Mn when $KMnO_4$ reacts with $H_2C_2O_4$ in acidic medium = y determines, (x + y)

CDF022

18. (a) Number of identical Cr–O bonds (having bond order more than 1) in $Cr_2O_7^{2\ominus}$ = x
(b) Number of identical Mn – O bonds in $MnO_4^{\ominus}$ = y determines, (x + y)

CDF023

The Lanthanoids (Electronic Configuration, Atomic and Ionic Sizes, Oxidation States)

19. Find the number of inner transition elements among those having following atomic numbers :
29, 59, 74, 95, 102, 104, 89, 57, 72.

CDF126

20. The number of unpaired electrons in $Gd^{3\oplus}$ is ..... .

CDF127

The Actinoids (Electronic Configuration, Atomic and Ionic Sizes, Oxidation States)

21. Most common oxidation state of actinoids is .... .

CDF125

For More Join: @IITJEE_Advanced

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Variable Oxidation States, Stability of higher O.S.

22. Out of the following how many oxides are acidic MnO, SO_3, V_2O_5, CrO_3, Mn_2O_7

CDF149

Trends in Standard electrode potentials

23. Find the number of elements from the given set have greater tendency to convert in M^{+2} to M^{+3} than Cr.
Ti, Fe, V, Mn, Co

CDF150

Magnetic Properties

24. Calculate spin only magnetic moment of divalent ion in aq. solution, if its atomic number is 25 (in BM)

CDF151

25. The spin only magnetic moment [in units of Bohr magneton] of Ni^{2+} in aqueous solution would be (At. No. of Ni = 28)

CDF152

Some Important Compounds of Transition Elements and their (Physical and Chemical) properties ($K_2Cr_2O_7$, K_2CrO_4, $KMnO_4$, K_2MnO_4)

26. When chromite ore is fused in strong alkali in presence of air, the number of equivalent bond lengths in anionic part of Cr-containing product is x. Find the value of $\frac{1}{x}$.

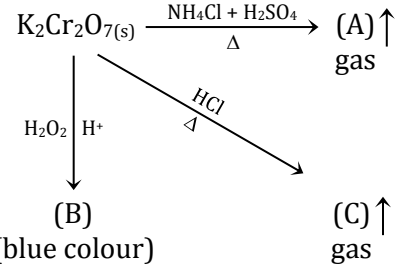
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CDF153

27. In fusion reaction of chromite ore with sodium carbonate in free access of air, 'a' number of elements undergo redox change & 'b' moles of O_2 are consumed per mole of chromite ore. Report (a + b)

CDF154

28.



Number of nonaxial d-orbital in hybridisation of central metal in gas A = x
Number of Cr – O bonds (sigma) in B = y
Molar mass of gas C = z
Find $\frac{Z}{X+Y}$ = ?

CDF155

29. In basic medium $K_2Cr_2O_7$ converts into a yellow coloured compound maximum change in oxidation state of chromium is:

CDF156

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. The spin only magnetic moment (μ) value (B.M.) of the compound with strongest oxidising power among Mn_2O_3, TiO and VO is _____ B.M. (Nearest integer). **[JEE (Main) 2025]**

CDF173

2. Match **List-I** with **List-II.** **[JEE (Main) 2025]**

List-I (Transition metal ion)	**List-II (Spin only magnetic moment (B.M.))**
(A) Ti^{3+}	(I) 3.87
(B) V^{2+}	(II) 0.00
(C) Ni^{2+}	(III) 1.73
(D) Sc^{3+}	(IV) 2.84

Choose the **correct** answer from the options given below :

(1) (A)-(III), (B)-(I), (C)-(II), (D)-(IV) (2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(3) (A)-(IV), (B)-(II), (C)-(III), (D)-(I) (4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

CDF174

3. The correct option with order of melting points of the pairs (Mn, Fe), (Tc, Ru) and (Re, Os) is :

(1) Fe < Mn, Ru < Tc and Re < Os (2) Mn < Fe, Tc < Ru and Re < Os

(3) Mn < Fe, Tc < Ru and Os < Re (4) Fe < Mn, Ru < Tc and Os < Re

[JEE (Main) 2025]

CDF175

4. Which of the following statements are correct about Zn, Cd and Hg?

A. They exhibit high enthalpy of atomization as the d-subshell is full.

B. Zn and Cd do not show variable oxidation state while Hg shows +I and +II.

C. Compounds of Zn, Cd and Hg are paramagnetic in nature.

D. Zn, Cd and Hg are called soft metals.

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Choose the most appropriate from the options given below: **[JEE (Main) 2024]**

(1) B, D only (2) B, C only (3) A, D only (4) C, D only

CDF157

5. The transition element that has lowest enthalpy of atomisation, is : **[JEE (Main) 2019]**

(1) Zn (2) Cu (3) V (4) Fe

CDF041

6. Which of the following ions is the strongest oxidizing agent ? **[JEE (Main) 2025]**

(Atomic Number of Ce = 58, Eu = 63, Tb = 65, Lu = 71)

(1) Lu^{3+} (2) Eu^{2+} (3) Tb^{4+} (4) Ce^{3+}

CDF176

7. The amphoteric oxide among V_2O_3, V_2O_4 and V_2O_5 upon reaction with alkali leads to formation of an oxide anion. The oxidation state of V in the oxide anion is : **[JEE (Main) 2025]**

(1) +3 (2) +7 (3) +5 (4) +4

CDF177

8. The set of correct statements is:

(i) Manganese exhibits +7 oxidation state in its oxide.

(ii) Ruthenium and Osmium exhibit +8 oxidation in their oxides.

(iii)Sc shows +4 oxidation state which is oxidizing in nature.

(iv) Cr shows oxidising nature in +6 oxidation state. **[JEE (Main) 2023]**

(1) (ii) and (iii) (2) (i), (ii) and (iv) (3) (i) and (iii) (4) (ii), (iii) and (iv)

CDF077

9. The correct order of basicity of oxides of vanadium is **[JEE (Main) 2023]**

(1) $V_2O_3 > V_2O_4 > V_2O_5$ (2) $V_2O_3 > V_2O_5 > V_2O_4$

(3) $V_2O_5 > V_2O_4 > V_2O_3$ (4) $V_2O_4 > V_2O_3 > V_2O_5$

CDF080

For More Join: @IITJEE_Advanced

10. The number of ions from the following that have the ability to liberate hydrogen from a dilute acid is _____. Ti^{2+}, Cr^{2+} and V^{2+} **[JEE (Main) 2024]**

(1) 0 (2) 2 (3) 3 (4) 1

CDF158

11. Given below are two statements : **[JEE (Main) 2024]**

Statement I : The higher oxidation states are more stable down the group among transition elements unlike p-block elements.

Statement II : Copper cannot liberate hydrogen from weak acids.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Statement I is true but Statement II is false

CDF159

12. Which of the following ions does not liberate hydrogen gas on reaction with dilute acids? **[JEE (Main) 2017]**

(1) $Ti^{2\oplus}$ (2) $Cr^{2\oplus}$ (3) $Mn^{2\oplus}$ (4) $V^{2\oplus}$

CDF038

13. Which of the following electronic configuration would be associated with the highest magnetic moment ? **[JEE (Main) 2024]**

(1) [Ar] $3d^7$ (2) [Ar] $3d^8$ (3) [Ar] $3d^3$ (4) [Ar] $3d^6$

CDF160

14. The spin-only magnetic moment value of $M^{3\oplus}$ ion (in gaseous state) from the pairs $Cr^{3\oplus}/Cr^{2\oplus}$, $Mn^{3\oplus}/Mn^{2\oplus}$, $Fe^{3\oplus}/Fe^{2\oplus}$ and $Co^{3\oplus}/Co^{2\oplus}$ that has negative standard electrode potential, is .....B.M. [Nearest integer] **[JEE (Main) 2022]**

Rankers Academy JEE

CDF047

15. Among $Co^{3\oplus}$, $Ti^{2\oplus}$, $V^{2\oplus}$ and $Cr^{2\oplus}$ ions, one if used as a reagent cannot liberate H_2 from dilute mineral acid solution, its spin-only magnetic moment in gaseous state is .....B.M. (Nearest integer) **[JEE (Main) 2022]**

CDF049

16. The statement that is **INCORRECT** about the interstitial compounds is : **[JEE (Main) 2019]**

(1) They have high melting points

(2) They are chemically reactive

(3) They have metallic conductivity

(4) They are very hard

CDF040

17. Match **List - I** with **List - II**. **[JEE (Main) 2025]**

List - I		List - II	
(A)	Bronze	(I)	Cu, Ni
(B)	Brass	(II)	Fe, Cr, Ni, C
(C)	UK silver coin	(III)	Cu, Zn
(D)	Stainless Steel	(IV)	Cu, Sn

Choose the **correct** answer from the options given below :

(1) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(2) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(3) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(4) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

CDF178

For More Join: @IITJEE_Advanced

18. Given below are two statements : **[JEE (Main) 2025]**

Statement I : In the oxalic acid vs $KMnO_4$ (in the presence of dil H_2SO_4) titration the solution needs to be heated initially to 60°C, but no heating is required in Ferrous ammonium sulphate (FAS) vs $KMnO_4$ titration (in the presence of dil H_2SO_4)

Statement II : In oxalic acid vs $KMnO_4$ titration, the initial formation of $MnSO_4$ takes place at high temperature, which then acts as catalyst for further reaction. In the case of FAS vs $KMnO_4$, heating oxidizes Fe^{2+} into Fe^{3-} by oxygen of air and error may be introduced in the experiment.

In the light of the above statements, choose the ***correct*** answer from the options given below :

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

CDF179

19. Which of the following oxidation reactions are carried out by both $K_2Cr_2O_7$ and $KMnO_4$ in acidic medium ? **[JEE (Main) 2025]**

A. $I^- \rightarrow I_2$

B. $S^{2-} \rightarrow S$

C. $Fe^{2+} \rightarrow Fe^{3+}$

D. $I^- \rightarrow IO_3^-$

E. $S_2O_3^{2-} \rightarrow SO_4^{2-}$

Choose the ***correct*** answer from the options given below :

(1) B, C and D only (2) A, D and E only

(3) A, B and C only (4) C, D and E only

CDF180

20. Consider the following reactions **[JEE (Main) 2025]**

$$K_2Cr_2O_7 \xrightarrow[-H_2O]{KOH} [A] \xrightarrow[-H_2O]{H_2SO_4} [B] + K_2SO_4$$

The products [A] and [B], respectively are :

(1) $K_2Cr(OH)_6$ and Cr_2O_3 (2) K_2CrO_4 and Cr_2O_3

(3) K_2CrO_4 and $K_2Cr_2O_7$ (4) K_2CrO_4 and CrO

CDF181

21. Preparation of potassium permanganate from MnO_2 involves two step process in which the 1st step is a reaction with KOH and KNO_3 to produce **[JEE (Main) 2025]**

(1) $K_4[Mn(OH)_6]$ (2) K_3MnO_4 (3) $KMnO_4$ (4) K_2MnO_4

CDF182

22. Consider the following reaction

$MnO_2 + KOH + O_2 \rightarrow A + H_2O$.

Product 'A' in neutral or acidic medium disproportionate to give products 'B' and 'C' along with water. The sum of spin-only magnetic moment values of B and C is _____ BM. (nearest integer)

(Given atomic number of Mn is 25) **[JEE (Main) 2024]**

CDF161

23. In alkaline medium. MnO_4^- oxidises I^- to **[JEE (Main) 2024]**

(1) IO_4^- (2) IO- (3) I_2 (4) IO_3^-

CDF162

24. Alkaline oxidative fusion of MnO_2 gives "A" which on electrolytic oxidation in alkaline solution produces B. A and B respectively are: **[JEE (Main) 2024]**

(1) Mn_2O_7 and MnO_4^- (2) MnO_4^{2-} and MnO_4^-

(3) Mn_2O_3 and MnO_4^{2-} (4) MnO_4^{2-} and Mn_2O_7

CDF163

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25. The fusion of chromite ore with sodium carbonate in the presence of air leads to the formation of products A and B along with the evolution of CO_2. The sum of spin-only magnetic moment values of A and B is ___ B.M. (Nearest integer)
(Given atomic number : C : 6, Na : 11, O : 8, Fe : 26, Cr : 24] **[JEE (Main) 2024]**

CDF164

26. Identify correct statements from below:
A. The chromate ion is square planar.
B. Dichromates are generally prepared from chromates.
C. The green manganate ion is diamagnetic.
D. Dark green coloured K_2MnO_4 disproportionate in a neutral or acidic medium to give permanganate.
E. With increasing oxidation number of transition metal, ionic character of the oxides decreases.
Choose the correct answer from the options given below: **[JEE (Main) 2024]**
(1) B, C, D only (2) A, D, E only (3) A, B, C only (4) B, D, E only

CDF165

27. The equilibrium $Cr_2O_7^{2-} \rightleftharpoons 2CrO_4^{2-}$ is shifted to the right in : **[JEE (Main) 2024]**
(1) an acidic medium (2) a basic medium
(3) a weakly acidic medium (4) a neutral medium

CDF166

28. Given below are two statements
Statement I : Aqueous solution of $K_2Cr_2O_7$ is preferred as a primary standard in volumetric analysis over $Na_2Cr_2O_7$ aqueous solution
Statement II : $K_2Cr_2O_7$ has a higher solubility in water than $Na_2Cr_2O_7$
In the light of the above statements, choose the correct answer from the options given below:
[JEE (Main) 2023]
(1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Both Statement I is true but Statement II is false
(4) Both Statement I is false but Statement II is true

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CDF167

29. Potassium dichromate acts as a strong oxidizing agent in acidic solution. During this process, the oxidation state changes from **[JEE (Main) 2023]**
(1) + 3 to + 1 (2) + 6 to + 3 (3) + 2 to + 1 (4) + 6 to + 2

CDF075

30. A solution of CrO_5 in amyl alcohol has a....colour **[JEE (Main) 2023]**
(1) Green (2) Orange–Red (3) Yellow (4) Blue

CDF078

31. The reaction of H_2O_2 with potassium permanganate in acidic medium leads to the formation of Mainly. **[JEE (Main) 2022]**
(1) $Mn^{2\oplus}$ (2) $Mn^{4\oplus}$ (3) $Mn^{3\oplus}$ (4) $Mn^{6\oplus}$

CDF048

32. Given below are two statements:
Statement I : Potassium permanganate on heating at 573 K forms potassium manganate.
Statement II : Both potassium permanganate and potassium manganate are tetrahedral and paramagnetic in nature.
In the light of the above statements, choose the most appropriate answer from the options given below: **[JEE (Main) 2021]**
(1) Statement I is true but statement II is false
(2) Both statement I and statement II are true
(3) Statement I is false but statement II is true
(4) Both statement I and statement II are false

CDF046

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33. Lanthanoid ions with $4f^7$ configuration are : **[JEE (Main) 2025]**

(A) Eu^{2+} (B) Gd^{3+} (C) Eu^{3+} (D) Tb^{3+}

(E) Sm^{2+}

Choose the correct answer from the options given below :

(1) (A) and (B) only (2) (A) and (D) only

(3) (B) and (E) only (4) (B) and (C) only

CDF183

34. Given below are two statements:

Statement (I) : In the Lanthanoids, the formation of Ce^{+4} is favoured by its noble gas configuration.

Statement (II) : Ce^{+4} is a strong oxidant reverting to the common +3 state.

In the light of the above statements, choose the most appropriate answer from the options given below: **[JEE (Main) 2024]**

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

CDF168

35. Which one amongst the following are good oxidizing agents?

A. $Sm^{2\oplus}$ B. $Ce^{2\oplus}$ C. $Ce^{4\oplus}$ D. $Tb^{4\oplus}$

Choose the most appropriate answer from the options given below : **[JEE (Main) 2023]**

(1) C only (2) D only (3) A and B only (4) C and D only

CDF050

36. The electronic configurations of bivalent Europium and trivalent Cerium are (atomic number : Xe = 54, Ce = 58, Eu = 63) **[JEE (Main) 2020]**

(1) [Xe] $4f^4$ and [Xe] $4f^9$ (2) [Xe] $4f^7$ and [Xe] $4f^1$

(3) [Xe] $4f^7$ $6s^2$ and [Xe] $4f^2$ $6s^2$ (4) [Xe] $4f^2$ and [Xe] $4f^7$

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CDF043

37. The correct order of atomic radii is : **[JEE (Main) 2019]**

(1) Ce > Eu > Ho > Nd (2) Nd > Ce > Eu > Ho

(3) Eu > Ce > Nd > Ho (4) Ho > Nd > Eu > Ce

CDF039

38. Number of colourless lanthanoid ions among the following is _____.

Eu^{3+}, Lu^{3+}, Nd^{3+}, La^{3+}, Sm^{3+} **[JEE (Main) 2024]**

CDF169

39. The number of element from the following that do not belong to lanthanoids is :

Eu, Cm, Er, Tb, Yb and Lu **[JEE (Main) 2024]**

(1) 3 (2) 4 (3) 1 (4) 5

CDF170

40. The maximum number of possible oxidation states of actinides are shown by **[JEE (Main) 2020]**

(1) Berkelium (Bk) and Californium (Cf)

(2) Nobelium (No) and Lawrencium (Lr)

(3) Actinium (Ac) and Thorium (Th)

(4) Neptunium (Np) and Plutonium (Pu)

CDF042

41. Mischmetal is an alloy consisting mainly of: **[JEE (Main) 2020]**

(1) Lanthanoid metals (2) Actinoid metals

(3) Actinoid and transition metals (4) Lanthanoid and Actinoid metals

CDF044

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EXERCISE - JEE (Advanced) PYQ

1. The correct statement(s) about $Cr^{2\oplus}$ and $Mn^{3\oplus}$ is (are)
[Atomic numbers of Cr = 24 and Mn = 25] **[JEE (Advanced) 2015]**

(A) $Cr^{2\oplus}$ is a reducing agent

(B) $Mn^{3\oplus}$ is an oxidizing agent

(C) Both $Cr^{2\oplus}$ and $Mn^{3\oplus}$ exhibit d^4 electronic configuration

(D) When $Cr^{2\oplus}$ is used as a reducing agent, the chromium ion attains d^5 electronic configuration

CDF071

2. An acidified solution of potassium chromate was layered with an equal volume of amyl alcohol. When it was shaken after the addition of 1 mL of 3% H_2O_2, a blue alcohol layer was obtained. The blue colour is due to the formation of a chromium (VI) compound **'X'**. What is the number of oxygen atoms bonded to chromium through only single bonds in a molecule of **X**?

[JEE (Advanced) 2020]

CDF073

3. Fusion of MnO_2 with KOH in presence of O_2 produces a salt W. Alkaline solution of W upon electrolytic oxidation yields another salt X. The manganese containing ions present in W and X, respectively, are Y and Z. Correct statement(s) is (are) **[JEE (Advanced) 2019]**

(A) Y is diamagnetic in nature while Z is paramagnetic

(B) Both Y and Z are coloured and have tetrahedral shape

(C) In both Y and Z, π-bonding occurs between p-orbitals of oxygen and d-orbitals of manganese.

(D) In aqueous acidic solution, Y undergoes disproportionation reaction to give Z and MnO_2.

CDF072

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	B	C	B	A	C	A	A	A
Que.	11	12	13	14	15	16				
Ans.	B	A	A	C	B	B				

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AB	AB	AB	AB	BC	ABC	BCD	ABC	ABC	D
Que.	11	12	13	14						
Ans.	B	C	A	D						

EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	8	9	8	3	3	2	3	3	14
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	5	3	6	4	3	4	8	10	3	7
Que.	21	22	23	24	25	26	27	28	29	
Ans.	3	3.00	0.00	5.92	2.84	0.25	4.75	8.87-8.88	0.00	

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	5	2	3	1	2	3	3	2	1	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	4	4	5	2	2	2	3	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	4	2	6	4	2	3	2	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	1	1	1	2	4	2	3	2	3	4
Que.	41									
Ans.	1									

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3
Ans.	ABC	4	BCD

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Important Notes

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